



NAUVAL ZULFIKAR

Data Scientist

Eligible to work in the UK (Graduate Visa)

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PROFILE

Results-driven Senior Data Scientist with 5+ years' experience, equipped with MSc in Business Analytics, building and deploying Machine Learning models, credit risk and fraud detection systems, ETL pipelines, and NLP solutions using Python, SQL, AWS, and A/B Testing, with deep expertise in regulated financial environments and data governance.

SKILLS

Python | SQL | R | PySpark | Hadoop | Docker | NoSQL | AI/ML | Deep Learning | NLP | LLM | RAG | Bayesian Statistics | Causal Inference | Time Series Analysis | Model Validation | Model Reliability | Credit Risk Modeling | Fraud Detection | Model Governance | Data Privacy | Workflow Orchestration | Model Robustness | Model Staleness | AI-powered Systems | Supervised Learning | Unsupervised Learning | Attribution Modeling | Customer Journey Optimisation | Web Analytics | A/B Testing | Predictive Modeling | Model Verification | Model Validation | Hypothesis Testing | Regression | Classification | AWS | ETL | Power BI | Tableau | Streamlit | Dashboards | Data Storytelling | NumPy | Scipy | scikit-learn | PyTorch | TensorFlow | Keras | Git

WORK EXPERIENCE

Public Works and Spatial Planning (PUTR) Department (Government)

Bandung, Indonesia

Data Scientist & Project Manager

September 2022 – Now

- **(Situation)** The department relied on manual data entry to update the region's building development and licensing, which delayed real-time insights delivery. **(Task)** I led a team to build data pipelines, automate processes, and design dashboards. **(Action)** I mentored junior members in building data pipelines, automated ETL processes, and designed self-service dashboards for stakeholders to access real-time data. **(Result)** The automation reduced project delivery time by 45%, improved data processing efficiency by 20%, and ensured data integrity and accessibility across analytics workflows — embedding governance and privacy practices into all pipeline processes, and empowered stakeholders with real-time insights for informed decision-making.
- **Tools:** Python, PySpark, AWS, Power BI, Social Media APIs, Agile Methodology, Jira.

Syncwell (Remote, Start-up)

Birmingham, United Kingdom

Marketing Data Scientist

July 2024 – September 2024

- **(Situation)** The company wanted to enhance marketing strategies by developing predictive models and improving customer engagement through data-driven insights. **(Task)** I was responsible for developing machine learning algorithms, improving customer segmentation, and driving engagement through campaign optimisation. **(Action)** I led the development of predictive machine learning algorithms, conducted A/B testing, and implemented automated campaign triggers and personalisation strategies. **(Result)** The efforts led to a 40% increase in click-through rates, 22% growth in followers, and improved customer journey.
- **Tools:** Python, SQL, Google Optimise, Social Media APIs, Power BI, Tableau, Salesforce.

PCOS Challenge (Remote, NPO)

Birmingham, United Kingdom

Strategic Data Scientist Specialist

June 2024 – August 2024

- **(Situation)** The organisation needed to build a system to track user engagement and optimise report generation. **(Task)** I led a team to develop a customer attribution model and automate reporting processes, aiming to reduce manual effort and enhance data visualisation for key insights. **(Action)** We developed a customer attribution model, automated report text extraction, and created interactive data visualisations to highlight patient financial concerns. **(Result)** The automation reduced manual workload by 80%, and the visualisations served better patient financial concerns insights.
- **Tools:** Python, Streamlit, Power BI, PySpark, Hadoop, Marketing Automation.

Bank Muamalat Indonesia (Corporation)

Jakarta, Indonesia

Lead Data Scientist – Leadership Development Program

November 2019 – August 2022

- **(Situation)** The task was to enhance portfolio analysis, improve marketing strategies, and support app development through data-driven insights. **(Task)** I led various data science initiatives, including developing NLP models, automating reporting, and applying clustering techniques for targeted marketing. **(Action)** I developed NLP models to optimise credit portfolio analysis and risk scoring, implemented real-time databases for cross-functional collaboration, led A/B testing to compare marketing strategies, and applied data governance and model validation practices in compliance with internal risk frameworks. Additionally, I utilised

clustering methods for market segmentation and designed an NLP-based app review analysis to compare the company's app with competitors. **(Result)** The NLP models increased investment returns by 5.3%, while marketing strategies became more replicable, targeting specific market segments and improving marketing personalisation. The app review analysis supported the app development division in refining the company's app features.

- Ensured model reliability and fairness through regular validation cycles, monitored model staleness, and maintained documentation aligned with regulatory and audit requirements in a licensed banking environment.
- **Tools:** AWS, Power BI, Docker, NoSQL, SQL, PySpark, Agile Methodology, Jira.

KEY PROJECTS AND PUBLICATIONS

Performance Analysis in Sport Footwear Sales Prediction Using Machine Learning January 2025
Publication

- Developed multiple Machine Learning models to be compared based on its accuracy.

Machine Scheduler Automation January 2025
Software Development Project

- **(Situation)** An advance machine scheduling system was required to automate and optimise production planning. **(Task)** I created a system that could efficiently schedule tasks, track machine utilisation, and allow for dynamic rescheduling. **(Action)** I developed a Streamlit-based machine scheduling system, designed interactive Gantt charts, and implemented real-time dashboards to visualise machine utilisation. **(Result)** The system successfully optimised production planning by 63% and minimised machines idle time.
- Tools: Python, Streamlit, **Workflow Orchestration**, Gantt Chart Visualization, Real-time Dashboards.

CV-Job Desc Suitability Checker Using Large Language Model November 2024
Software Development Project

- **(Situation)** Due to the disadvantageous of current Applicant Tracking System (ATS) system, an automated job matching tool to streamline recruitment processes and improve candidate-job relevance. **(Task)** I developed a system that could assess the suitability of CVs for job descriptions and outperform traditional ATS benchmarks. **(Action)** I created a DeBERTa-v3-based job matching tool, incorporated RAG-based retrieval, and designed reproducible ML pipelines with end-to-end orchestration to ensure scalability and maintainability. **(Result)** The system achieved higher relevance than traditional ATS methods, improving the accuracy and efficiency of the recruitment automation process.

FORMAL EDUCATION

Aston University (1:1 Acquired, EQF Level 7) **Birmingham, United Kingdom**
MSc Business Analytics September 2023 – September 2024

- 1st Winner of the **Aston Enterprise Scholarship 2023** awarded a full-tuition scholarship.
- **Aston Pitching Contest 2024** runner-up for presenting my business idea in AI-powered supply chain tools
- Dissertation: “**Enhancing Supply Chain Information Sharing Systems Through Blockchain**”
- **Distinctions in:** Effective Management Consultancy, Decision Models, Software Analytics

Ritsumeikan Asia Pacific University (EQF Level 6) **Beppu, Japan**
BBA Innovation and Economics October 2016 – September 2019

- Recipient of **APU 50% Tuition Reduction Scholarship** and **Japan Student Services Organisation (JASSO) Scholarship**
- **Distinctions in:** Business Data Analysis, Consumer Behaviour, Marketing Research

INFORMAL EDUCATION

Qarir Generator (Bootcamp, EQF Level 5) **Dubai, UAE**
Data Science Global Accelerator April 2023 – June 2023

- Led the development of sentiment analysis process for **Pfizer Public Sentiment Analysis** project
- **Distinctions in:** Python, SQL, and Pandas/NumPy

Indonesia AI (Bootcamp, EQF Level 5) **Indonesia**
Natural Language Processing (NLP) Engineer Training May 2023 – November 2023

- Contributed to the development of **2024 Indonesia Election Aspect-Based Sentiment Analyser** project
- **Distinctions in:** Python Programming, Text Summerisation, and Voice Recognition

LANGUAGE

- **Indonesian:** Reading (Native), Listening (Native), Writing (Native), Speaking (Native)
- **English:** Reading (Native), Listening (Native), Writing (Native), Speaking (Native)
- **Japanese:** Reading (Intermediate), Listening (Intermediate), Writing (Intermediate), Speaking (Intermediate)